

Data 2

1. $\bar{w} = 3.66$

$$\sigma_w = 1.03$$

2. a.

$$\begin{aligned}\sigma^2_m &= \frac{\sum m^2}{7} - \left(\frac{\sum m}{7}\right)^2 \\ &= \frac{9 + 16 + 36 + 4 + 64 + 64 + 25}{7} - \left(\frac{3 + 4 + 6 + 2 + 8 + 8 + 5}{7}\right)^2 \\ &= 4.69\end{aligned}$$

$$\therefore \sigma_m = 2.17$$

b. $\sigma^2_m = 4.69$

$$\sigma_m = 2.17$$

3. a. On the x-axis, “percentage of original value”. On the y-axis, “frequency density”.

b. $\bar{v} = 29.9$. $\sigma_v = 7.54$

Outliers are more than 2σ away from $\bar{v}$, so less than $29.9 - 2(7.54) = 14.82$ or greater than $29.9 + 2(7.54) = 44.98$.

So there are two outliers which are greater than 45% of original value.

c. The price of a five-year-old diesel car is, on average, greater than the price of a five-year-old petrol car. However, because the standard deviation in the similar survey is higher, the price of the diesel car varies more after 5 years than the price of a petrol car.

4. a. median = 42.5. Lower quartile = 40.5. Upper quartile = 43.3.

b.

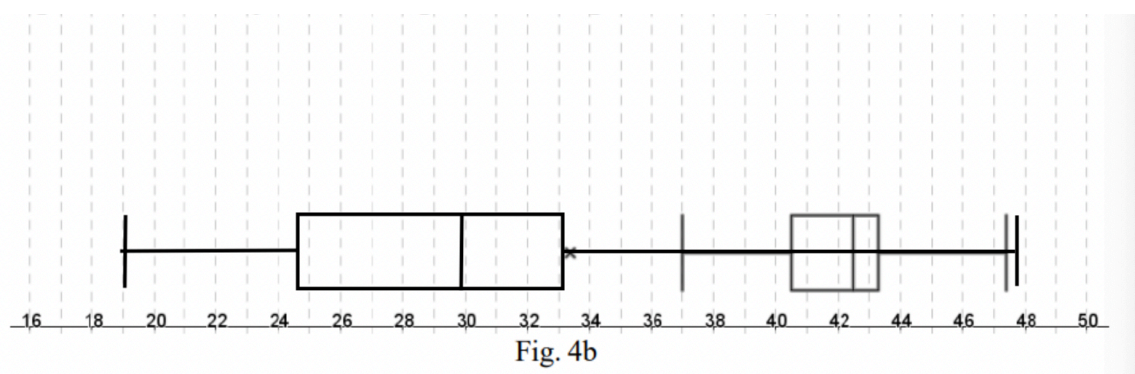
$$IQR = 2.8$$

$$1.5 \times IQR = 4.2$$

So outliers are ≤ 36.3 or ≥ 47.5 . 33.4 is the only value that fits into this definition, so it is the only outlier.

c. Here, table 4a has $IQR = 8.6$, so outliers are ≥ 41.7 or ≤ 15.9 . The only country that has a median age above the outlier is Japan, so Japan is the only outlier.

d.



e. European countries have a much lower mean age than Asian countries, but the range of European ages is much larger than the range of Asian countries' ages.

f. Mean = $\frac{1048.3}{35} = 30.0$.

$$\text{Standard deviation} = \sqrt{\left(\frac{32942.71}{35}\right) - \left(\frac{1048.3}{35}\right)^2} = 6.64.$$